

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination 2024

Course: B.Tech **Branch :** Computer Engg/Computer Science & Engg **Semester:** VI

Subject Code & Name: Computer Networks BTCC602

Max Marks: 60

Date: 20/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The is responsible for moving individual bits from one node to the next.	understand	1
	a. Data link b. Physical c. Transport d. Network		
2	ISO is the organization; OSI is the in Computer Network.	understand	1
	a. Network b. Domain c. Model d. Method		
3	The X.25 is a	understand	1
	a. Topology b. Protocol c. Network d. Routing Tech.		
4	A Bluetooth network is called a	understand	1
	a. Ethernet b. supernet c. subnet d. piconet		
5	The unit of communication at the data link layer is a	understand	1
	a. segment b. datagram c. Frame d. packet		
6	Which of the following tasks is not done by data link layer?	Analyze	1
	a. flow control b. error control c. chan. coding d. framing		
7	A 4 byte IP address consists of	understand	1
	a. only network address b. only host address c. network add. & host address d. network add. & MAC address		

8	A _____ is a TCP name for a transport service access point.				understand	1
	a. pipe	b. port	c. node	d. protocol		
9	The domain name system is maintained by _____				analyze	1
	a. a single server	b. a single computer	c. distributed database sys.	d. none of the mentioned		
10	FTP uses _____ parallel TCP connections to transfer a file.				understand	1
	a. 1	b. 3	c. 2	d. 4		
11	Network layer firewall works as a _____				understand	1
	a. Frame filter	b. Packet filter	c. Content filter	d. Virus filter		
12	Which is not an objective of network security?				analyze	1
	a. Identification	b. Authentication	c. Access control	d. Lock		
Q. 2	Solve the following.					12
A)	Explain functions of various layers in TCP/IP model in brief.				understand	6
B)	Draw Ethernet frame format and explain function of each field.				analyze	6
Q. 3	Solve the following.					12
A)	Given the dataword 1101101 and the divisor is 10101, generate the cyclic redundancy check (CRC) codeword at the sender site.				Apply	6
B)	Draw UDP header format and explain its working				understand	6
Q. 4	Solve Any Two of the following.					12
A)	Illustrate byte count and bit stuffing framing techniques .				understand	6
B)	Explain ipv4 classful addressing in detail.				Understand	6
C)	Describe various congestion control policies of TCP in brief.				Analyze	6
Q. 5	Solve Any Two of the following.					12
A)	Compare SMTP and POP protocols.				Analyze	6
B)	Describe working of FTP in detail.				Understand	6
C)	Explain Distance vector routing with suitable example.				analyze	6

Q. 6	Solve Any Two of the following.		12
A)	The following is a dump of a TCP header in hexadecimal format 053200217 000000001 00000000 500207FF 00000000 1) What is the source port number? 2) What is the destination port number? 3) What is sequence number? 4) What is the acknowledgement number? 5) What is the length of the header? 6) What is the window size?	apply	6
B)	Illustrate with example public key and private key cryptography	understand	6
C)	Explain design issues of data link layer.	understand	6
	*** End ***		